$\hat{y} = m_1 x_1 + m_2 x_2 + b$

Initial values

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

learning rate = 0.01

Data points.

$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$ $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

learning rate = 0.01.

1. Computing the predicted $\hat{y}$ for each data Point.

$\hat{y} = m_1 x_1 + m_2 x_2 + b$

$= (-1)(1) + (2)(3) + 1 = -1 + 6 + 1 = 6$

$\hat{y} = (-1)(4) + (2)(10) + 1 = -4 + 20 + 1 = 17$

$\hat{y} = [6, 17]$

2. The gradient of MSE

$e = \hat{y} - y = m_1 x_1 + m_2 x_2 + b - y$

$MSE = \frac{1}{n} \Sigma (\hat{y} - y)^2 = \frac{1^2 + 11^2}{2} = \frac{122}{2} = 61$

$e_1 = 6 - 5 = 1$

$e_2 = 17 - 6 = 11$

$m_{new} = m_{old} - \alpha \frac{\partial MSE}{\partial m}$

$b_{new} = b_{old} - \alpha \frac{\partial MSE}{\partial b}$

$\frac{\partial MSE}{\partial m} = \frac{1}{n} \frac{\partial (\hat{y} - y)^2}{\partial m} = \frac{\partial (m_1 x_1 + m_2 x_2 + b - y)^2}{\partial m}$

With respect to m_1

$\frac{\partial (m_1 x_1 + m_2 x_2 + b - y)^2}{\partial m_1}$ let $m_2 x_2 + b - y = c$

$\frac{\partial (m_1 x_1 + c)^2}{\partial m}$

$m_1 x_1 + c = u$

$\therefore \frac{dy}{dm_1} = x_1$

$MSE = u^2$

$\frac{d \cdot MSE}{du} = 2u$

Chain rule

$$\frac{\partial mse}{\partial m_1} = \frac{\partial mse}{\partial u} \cdot \frac{\partial u}{\partial m_1} = 2u \cdot x_1$$

$$= 2(m_1 x_1 + m_2 x_2 + b - y) x_1$$

$$= 2e x_1 \qquad = \frac{2}{n} \Sigma e x_1$$

With Respect to m_2

$$\frac{\partial mse}{\partial m_2} = \frac{\partial (\hat{y} - y)^2}{\partial m_2} = \frac{\partial (m_1 x_1 + m_2 x_2 + y + b)^2}{\partial m_2}$$

let $m_1 x_1 + b - y = c$

$$= \frac{\partial (m_2 x_2 + c)^2}{\partial m_2}$$

let $m_2 x_2 + c = u$ $\qquad mse = u^2$

$\frac{du}{dm_2} = x_2$ $\qquad \frac{dmse}{du} = 2u$

Chain Rule

$$\frac{\partial mse}{\partial m_2} = \frac{\partial mse}{\partial u} \cdot \frac{\partial u}{\partial m_2} = 2u \cdot x_2$$

$$= 2(m_1 x_1 + m_2 x_2 + b - y) x_2 = 2e x_2 = \frac{2}{n} \Sigma e x_2$$

$$\frac{\partial mse}{\partial b} = \frac{\partial (\hat{y} - y)^2}{\partial b} = \frac{\partial (m_1 x_1 + m_2 x_2 + b - y)^2}{\partial b}$$

let $m_1 x_1 + m_2 x_2 - y = c$

$$\frac{\partial (b + c)^2}{\partial b}$$

let $b + c = u$. $\qquad mse = u^2$

$\frac{du}{db} = 1$ $\qquad \frac{dmse}{du} = 2u$.

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Chain Rule:

$$\frac{dmse}{db} = \frac{\partial mse}{\partial u} \cdot \frac{\partial u}{\partial b} = 2u \cdot 1$$

$$= 2(m_1 x_1 + m_2 x_2 + b - y) \cdot$$

$$= \frac{2}{n} e = \frac{2}{n} (e)$$

$$ex_1 = (1)(1) + 4(11) = 1 + 44 = 45$$

$$ex_2 = (1)(3) + (11)(10) = 3 + 110 = 113.$$

1st Iteration

$$m_{new} = m_{old} - \alpha \frac{\partial mse}{\partial m}$$

$$m_{1\ new} = -1 - 0.01(45) = -1 - 0.45 = -1.45$$

$$m_{2\ new} = 2 - 0.01(113) = 2 - 1.18 = 0.87$$

$$M = \begin{bmatrix} -1.45 \\ +0.87 \end{bmatrix}$$

$$b_{new} = b_{old} - \alpha \frac{\partial mse}{\partial b}$$

$$b_{1\ new} = 1 - 0.01(1) = 0.99$$

$$b_{2\ new} = 1 - 0.01(11) = 0.89$$

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2nd Iteration

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \qquad b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} \qquad x = ((1, 3)(4, 10)) \qquad y = (5, 6)$$

$$\hat{y} = m_1 x_1 + m_2 x_2 + b$$

$$= (-1.45)(1) + (0.87)(3) + 0.99 = 2.15$$

$$= (-1.45)(4) + (0.87)(10) + 0.89 = 3.79$$

$$\hat{y} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$e = \hat{y} - y$$

$$= \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$MSE = \frac{1}{n} \cdot \Sigma (\hat{y} - y)^2$$

$$= \frac{1}{2}\left((-2.85)^2 + (-2.21)^2\right)$$

$$= 6.5033$$

$$\frac{dMSE}{dm_1} = e x_1$$

$$= (-2.85)(1) + (-2.21)(4)$$

$$= -11.69$$

$$\frac{dMSE}{dm_2} = e x_2$$

$$= (-2.85)(3) + (-2.21)(10)$$

$$= -30.65$$

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$$\frac{dMSE}{db} = e$$

$$\frac{dMSE}{db_1} = e_1$$

$$\frac{dMSE}{db_1} = -2.85$$

$$\frac{dMSE}{db_2} = -2.21$$

$$m_{new} = m_{old} - \alpha \frac{dMSE}{dm}$$

$$m_{1new} = -1.45 - (0.01 \times -11.69)$$

$$= -1.3331$$

$$m_{2new} = 0.87 - (0.01 \times -30.65)$$

$$= 1.1765$$

$$b_{new} = b_{old} - \alpha \frac{dMSE}{db}$$

$$b_{1new} = 0.99 - (0.01 \times -2.85)$$

$$= 1.0185$$

$$b_{2new} = 0.89 - (0.01 \times -2.21)$$

$$= 0.9121$$

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$m = [-1.3331, 1.1765]$ (m_1, m_2) $\quad b = [1.0185, 0.9121]$ (b_1, b_2)

$\hat{y} = m_1 x_1 + m_2 x_2 + b$

$\hat{y}_1 = (-1.3331)(1) + (1.1765)(3) + 1.0185$

$= -1.3331 + 3.5295 + 1.0185$

$\hat{y}_1 = 3.2149$

$\hat{y}_2 = (-1.3331)(4) + (1.1765)(10) + 0.9121$

$= -5.3324 + 11.765 + 0.912$

$\hat{y}_2 = 7.3447$

$\text{Error}_{(e)} = y - \hat{y}$

$e_1 = 3.2149 - 5 = -1.7851$

$e_2 = 7.3447 - 6 = 1.3447$

$MSE = \frac{1}{n}(e_1^2 + e_2^2) = \frac{1}{2}\left((-1.7851)^2 + (1.3447)^2\right)$

$= \frac{1}{2}(3.1865 + 1.8082)$

$= \frac{1}{2}(4.9948) = 2.4974$

Chain rule

$MSE = \frac{1}{n}\sum e_i^2$

$\frac{\partial MSE}{\partial e_i} = \frac{2}{n} e_i = \frac{2}{2} e_i$

$\therefore \frac{dMSE}{de_1} = e_1$

$\frac{dMSE}{de_2} = e_2$

~~$\hat{y} = m_1 x_1 + m_2 x_2 + b$~~ $MSE = \frac{1}{n}(\hat{y} - y)^2$

$\frac{dMSE}{dm} = \frac{1}{n}\frac{(\hat{y} - y)^2}{dm} = \frac{1}{n}\frac{(m_1 x_1 + m_2 x_2 + b - y)^2}{dm}$

$\therefore \frac{dMSE}{dm_1} = \frac{d(m_1 x_1 + m_2 x_2 + b - y)^2}{dm_1}$ Let $m_2 x_2 + b - y = c$ then be constant

$= \frac{d(m_1 x_1 + c)}{dm_1}$ Let $m_1 x_1 + c = u$

$\therefore MSE = u^2$

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$$\frac{dy}{dm_1} = x_1 \qquad \frac{dmse}{du} = 2u$$

$$\frac{dmse}{dm_1} = \frac{dmse}{du} \cdot \frac{du}{dm_1}$$

$$= 2u \cdot x_1$$

Replace $u = [2(m_1x_1 + m_2x_2 + b - y)]x_1$

$$= 2(e)x_1 = \frac{2}{n} \varepsilon x_1$$

$$\frac{dmse}{db} = \frac{d(\hat{y}-y)^2}{db} = \frac{d(m_1x_1 + m_2x_2 + b - y)^2}{db}$$

Let $m_1x_1 + m_2x_2 - y = c \Rightarrow \frac{d(b+c)^2}{db}$

Let $b + c = u$
$mse = u^2 \Rightarrow \frac{du}{db} = 1 \qquad \frac{dmse}{du} = 2u$

$$\frac{dmse}{db} = \frac{dmse}{du} \cdot \frac{du}{db}$$

$$= 2u \cdot 1$$

$$= 2(m_1x_1 + m_2x_2 + b - y)$$

$$= \frac{2}{n}(e)$$

$$\frac{dmse}{dm_2} = \frac{d(\hat{y}-y)^2}{dm_2} = \frac{d(m_1x_1 + m_2x_2 + b - y)^2}{dm_2}$$

Let $m_1x_1 + b - y = c \quad = \frac{d(m_2x_2 + c)^2}{dm_2}$

Let $m_2x_2 + c = u \quad = \frac{d(u)^2}{dm_2}$

$\therefore mse = u^2$

$$\frac{du}{dm_2} = x_2 \qquad \frac{dmse}{du} = 2u$$

$$\frac{dmse}{dm_2} = \frac{dmse}{du} \cdot \frac{du}{dm_2} = 2u \cdot x_2$$

$$= [2(m_2x_2 + m_1x_1 + b - y)]x_2$$

$$= 2ex_2 = \frac{2}{n} \varepsilon x_2$$

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$$\frac{dmse}{dm_1} = (-1.7851)(1) + (1.3447)(4) = -1.7851 + 5.3788$$

$$= \underline{3.5937}$$

$$\frac{dmse}{dm_2} = (-1.7851)(3) + (1.3447)(10) = -5.3553 + 13.4470$$

$$= \underline{8.01917}$$

$$\frac{dmse}{db_1} = \underline{-1.7851} \qquad \frac{dmse}{db_2} = \underline{1.3447}$$

3rd iteration

$$m_1 new = m_{old} - \alpha \frac{dmse}{dm_1}$$

$$= -1.3331 - (0.01)(3.5937)$$

$$= -1.3331 - 0.03597$$

$$m_1 new = \underline{-1.369037}$$

$$m_2 new = m_{old} - \alpha \frac{dmse}{dm_2}$$

$$= 1.1765 - (0.01)(8.0917)$$

$$= 1.1765 - 0.080917$$

$$m_2 new = \underline{1.095583}$$

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$$b_{1\,new} = b_{old} - \alpha \frac{dmse}{db_1}$$

$$= 1.0185 - (0.01)(-1.7851)$$

$$= 1.0185 + 0.017851$$

$$b_{1\,new} = \underline{1.036351}$$

$$b_{2\,new} = b_{old} - \alpha \frac{dmse}{db_2}$$

$$= 0.9121 - (0.01)(1.3447)$$

$$= 0.9121 - 0.013447$$

$$b_{2\,new} = \underline{0.898653}$$

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